Factor Analysis of Mental Ability Test Scores

Case Study

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Multivariate Data Analysis

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Abstract:

This report presents an exploratory factor analysis of nine mental ability test scores from the Holzinger-Swineford 1939 dataset, comprising 301 students across two schools. The objective is to identify the latent dimensions underlying the observed test correlations, evaluate the adequacy of a three-factor solution, and use factor scores to compare student groups by school, grade, and sex.

Descriptive statistics, correlation diagnostics (KMO = 0.75; Bartlett's test $p < 0.001$), and a scree plot supported the suitability of factor analysis. Information criteria (AIC/BIC) modestly favoured a two-factor solution, but theoretical interpretability and variance explained supported three factors. A three-factor maximum likelihood solution explained 54% of the total variance, with a residual Frobenius norm of 0.163, indicating acceptable model fit. Varimax and promax rotations were compared, with the oblique rotation allowing for modest factor correlations. The three factors were interpreted as (1) visual–spatial ability, (2) textual–verbal ability, and (3) processing speed. Factor scores revealed meaningful group differences: Pasteur students scored higher on verbal ability, while Grant-White students scored higher on spatial ability; eighth graders consistently outperformed seventh graders; and small sex differences were observed in processing speed and verbal ability.

The results demonstrate that a three-factor structure parsimoniously summarises the correlation pattern and provides interpretable dimensions for educational assessment.

Keywords: Factor analysis, mental ability, psychometrics, rotation, factor scores, group comparison

1. Introduction

Psychologists and educational measurement specialists frequently use multiple test scores to measure broad latent abilities that cannot be observed directly. For example, several spatial tasks may reflect an underlying visual–spatial factor, language tasks may reflect a verbal factor, and timed tasks may reflect a processing-speed factor. Because observed test scores are correlated, factor analysis can identify a smaller number of latent dimensions that summarise the pattern of student performance.

In this case study, the role is that of a statistician supporting an educational assessment team. The primary objective is to apply exploratory factor analysis to the classic Holzinger-Swineford (1939) mental ability dataset, which contains scores from students at two schools (Pasteur and Grant-White). Specifically, the analysis aims to: (i) determine how many latent ability dimensions are supported by the data; (ii) interpret those dimensions using rotated factor loadings; (iii) evaluate whether the chosen factor model adequately reproduces the observed correlation matrix; and (iv) compute factor scores to compare students across schools, grades, and sex.

2. Materials / Data

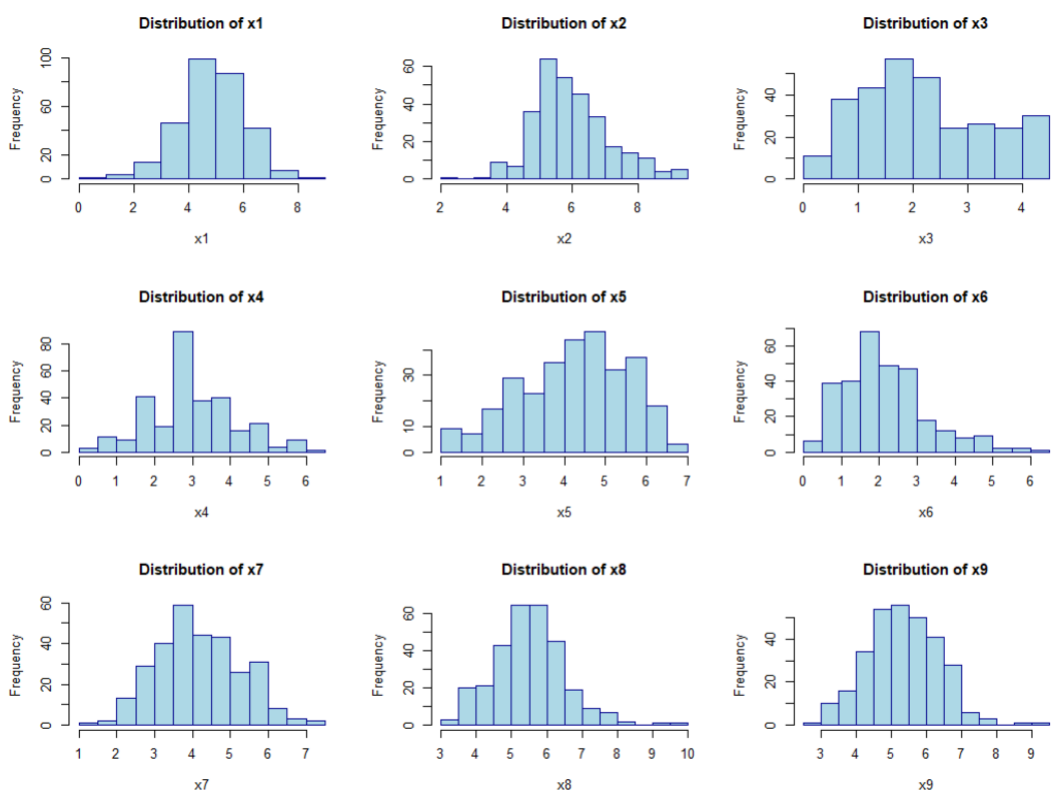
The data are from the Holzinger-Swineford (1939) mental ability study, a classic dataset in psychometrics and factor analysis. In the version commonly distributed with the R lavaan package, the dataset contains 301 observations and 15 variables. The unit of analysis is individual students enrolled at two schools: Pasteur and Grant-White.

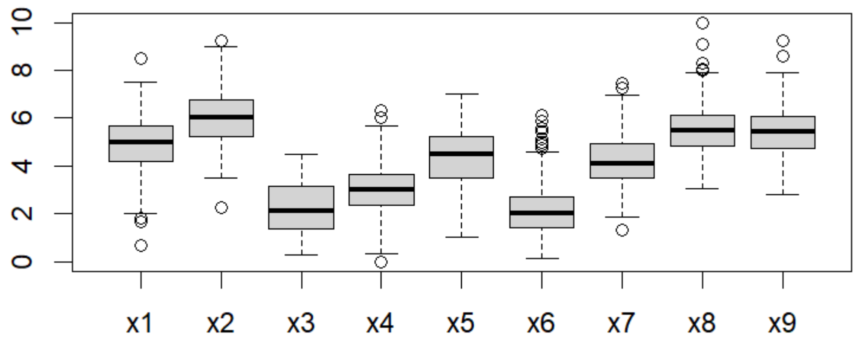
The analysis focuses on the nine continuous test-score variables, x1 through x9, measured on approximate continuous scales. These variables are designed to capture three broad domains (visual-spatial, verbal, and processing speed), though the number of latent factors was determined empirically rather than assumed in advance.

Variable	Test Description	Expected Domain to consider
x1	Visual perception	Visual / Spatial ability
x2	Cubes	Visual / Spatial ability

x3	Lozenges	Visual / Spatial ability
x4	Paragraph comprehension	Textual / Verbal ability
x5	Sentence completion	Textual / Verbal ability
x6	Word meaning	Textual / Verbal ability
x7	Speeded addition	Processing speed
x8	Speeded counting of dots	Processing speed
x9	Speeded discrimination of straight and curved capitals	Processing speed

A working dataset was created containing only x1 through x9 (n = 301). Descriptive statistics were computed, including means, standard deviations, minimum, maximum, skewness, and kurtosis. Histograms and boxplots were examined to assess approximate normality and to detect potential outliers.





3. Methods

All statistical analyses were performed using R (Version 4.4). The primary packages used were psych for factor analysis diagnostics, GPArotation for factor rotation, and rgl for 3D visualisation of loadings. A significance level of $\alpha = 0.01$ was adopted throughout.

Data Preparation

The nine continuous test-score variables (x1 to x9) were extracted from the full dataset to form a working dataframe (df). Descriptive statistics (mean, standard deviation, minimum, maximum, skewness, kurtosis) were computed for each variable. Histograms and boxplots were generated to assess approximate normality and to identify potential outliers.

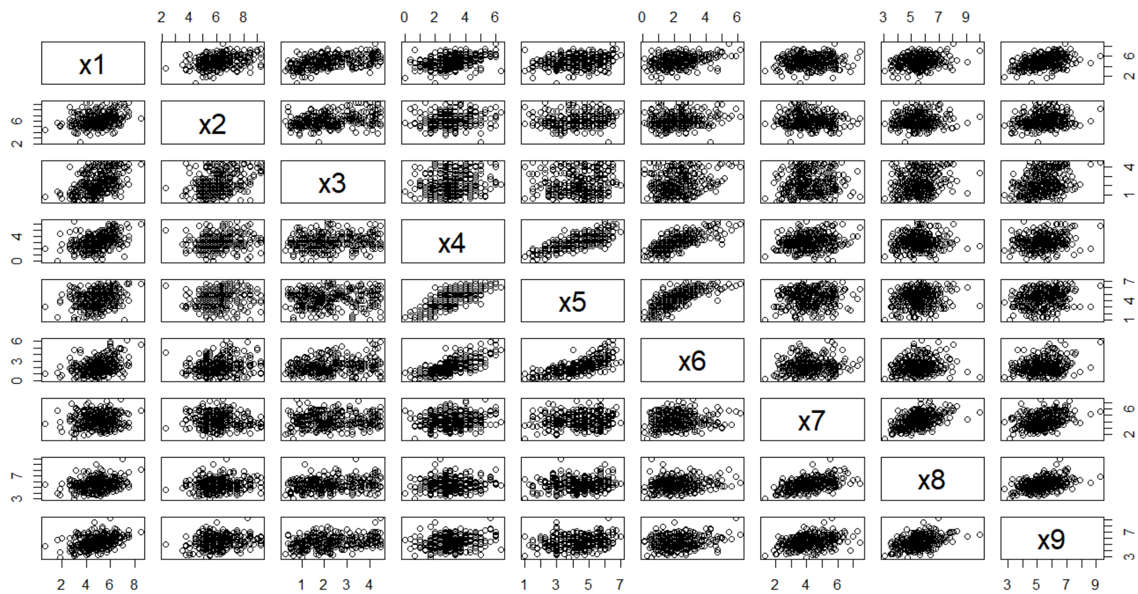
Factorability of the Correlation Matrix

Before proceeding with factor analysis, the suitability of the data for dimension reduction was assessed using two standard diagnostics:

- **Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy:** Values above 0.60 are considered acceptable, with values above 0.80 considered good. The KMO indicates whether the variables share sufficient common variance for factor analysis.

- **Bartlett's test of sphericity:** This tests the null hypothesis that the correlation matrix is an identity matrix (i.e., all variables are uncorrelated). A significant test (p < 0.01) indicates that the correlations are sufficiently large to proceed with factor analysis.

The correlation matrix of the nine variables was also inspected visually using a pairs plot and numerically using the cor() function.



Determining the Number of Factors

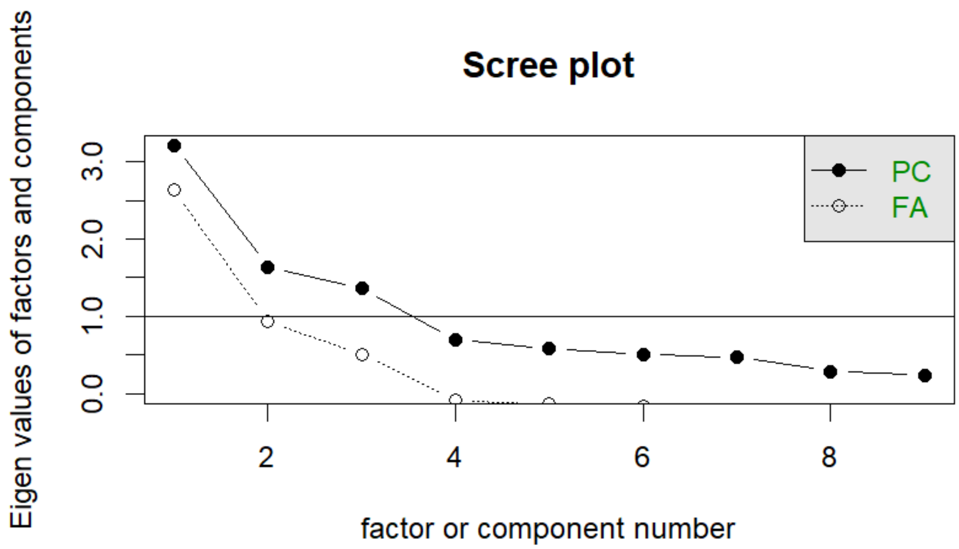
The number of factors to retain was determined using multiple criteria to avoid reliance on any single heuristic:

- **Scree plot (Cattell, 1966):** Eigenvalues were plotted against factor number. The "elbow" of the scree plot indicates the point where additional factors explain little additional variance.

- **Kaiser's criterion:** Factors with eigenvalues greater than 1 were considered for retention.

- **Variance explained:** The cumulative proportion of variance explained by candidate factor solutions (e.g., 2-factor, 3-factor, 4-factor) was compared.

- **Expert knowledge:** Based on the expected domain structure (visual-spatial, verbal, processing speed), a three-factor solution was considered substantively plausible.



Factor Extraction

Factor analysis was performed using maximum likelihood (ML) estimation via the factanal() function in R. ML estimation was chosen because it provides:

- A formal goodness-of-fit test for the number of factors;
- Standard errors and confidence intervals for loadings;
- Asymptotically efficient estimates under multivariate normality assumptions.

Both three-factor (fa3) and four-factor (fa4) models were fitted to compare model fit and parsimony. The following statistics were extracted from each model:

- Factor loadings: Correlations between observed variables and latent factors, interpreted as the contribution of each factor to each variable.
- Communalities **(h^2):** The proportion of variance in each observed variable explained by the retained factors (calculated as the row-sum of squared loadings).
- Uniquenesses **(u^2):** The proportion of variance not explained by the factors (calculated as 1 − communality).
- Proportion of total variance explained: The sum of squared loadings for each factor divided by the number of variables.

- Goodness-of-fit test: The null hypothesis that the retained number of factors is sufficient was tested using the chi-square statistic provided by factanal(). A non-significant p-value ($p \geq 0.05$) indicates that the model fits the data adequately; a significant p-value suggests that more factors may be needed.

Model Adequacy: Residual Analysis

To evaluate how well the chosen factor model reproduced the observed correlations, the residual correlation matrix was computed as:

$$\mathbf{R}_{\text{residual}} = \mathbf{R}_{\text{observed}} - \left(\mathbf{\Lambda}\mathbf{\Lambda}^{\top} + \mathbf{\Psi}\right)$$

where $\mathbf{\Lambda}$ is the matrix of factor loadings and $\mathbf{\Psi}$ is the diagonal matrix of uniquenesses. The overall magnitude of the residuals was summarised using the Frobenius norm:

$$\|\mathbf{D}\|_F = \sqrt{\sum_{i=1}^{p}\sum_{j=1}^{p} d_{ij}^2}$$

where dij are the residual correlations. Smaller values indicate better model fit.

Factor Rotation

To aid interpretation, the factor solution was rotated using both orthogonal and oblique methods:

- Varimax rotation (orthogonal): Assumes factors are uncorrelated. This rotation simplifies loadings by maximising the variance of squared loadings within each factor.

- Promax rotation (oblique): Allows factors to be correlated. Because cognitive abilities may plausibly be correlated (e.g., students with high verbal ability may also have higher processing speed), an oblique solution may be more realistic.

Promax was implemented as it is a standard, computationally efficient oblique rotation.

Both rotated solutions were fitted using factanal() with the rotation = "varimax" and rotation = "promax" arguments. For the oblique solution, the factor correlation matrix was examined to assess the magnitude and direction of associations among latent factors.

Visualisation: Three-dimensional scatterplots of the unrotated, varimax-rotated, and promax-rotated loadings were generated using the rgl package to visualise the effect of rotation on factor structure.

Factor Scores

Factor scores were estimated using the principal component method (principal() function in the psych package) rather than the default regression method. This approach was chosen for computational simplicity and because it produces scores with mean zero and unit variances. Specifically, scores were computed as:

$$\mathbf{F} = \mathbf{Z}\,\mathbf{R}^{-1}\boldsymbol{\Lambda}$$

where **Z** is the matrix of standardised observed scores, **R^(-1)** is the inverse of the correlation matrix, and **Λ** is the matrix of factor loadings from the principal component solution. The resulting scores were added to the original dataset as three new variables (pc1, pc2, pc3).

For comparison and consistency, the maximum likelihood model score estimations were calculated as well. The R function factanal() was used, and the estimation of the scores was via vanilla regression. The scores were numerically different from the pca scores, which is expected. The interpretation and relationships between scores (signs, magnitude of differences, etc) were consistent across the two methods.

Group Comparisons

Mean factor scores were compared across three student subgroups using descriptive statistics (means rounded to two decimal places):

- School: Pasteur vs. Grant-White
- Grade: 7 vs. 8
- Sex: Male (sex = 1) vs. Female (sex = 2)

Comparisons were performed by subsetting the complete dataset (df.complete) and applying the colMeans() function to the factor score columns. No formal inferential tests (e.g., t-tests or ANOVA) were conducted, as the assignment required only descriptive group comparisons. However, the observed differences are reported as descriptive patterns to be interpreted cautiously.

4. Results

The correlation matrix (Table 1) showed moderate to strong correlations within expected domains. Variables x4, x5, and x6 (verbal tasks) correlated strongly with each other ($r = 0.70$–0.73). Variables x1, x2, x3 (spatial tasks) showed weaker but positive correlations ($r = 0.30$–0.44). Variables x7, x8, x9 (speeded tasks) correlated moderately with each other ($r = 0.45$–0.49).

Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy: Overall MSA = 0.75 (meritorious). Individual MSAs ranged from 0.59 (x7) to 0.81 (x1, x6). Values above 0.60 are acceptable; x7 (0.59) was marginal but retained due to theoretical relevance.

Bartlett's test of sphericity: $\chi^2(36) = 904.10$, $p < 0.001$. The null hypothesis of an identity correlation matrix was rejected. The data are suitable for factor analysis.

Number of Factors

A scree plot of eigenvalues showed a clear elbow between 2 and 3 factors. Kaiser's criterion (eigenvalue > 1) suggested 1 factor (eigenvalue = 3.22). However, the elbow of the scree plot and cumulative variance explained supported 2–3 factors.

Based on the scree plot elbow (see methods section), the interpretability of the three-factor solution, and the expected domain structure (visual-spatial, verbal, processing speed), three factors were retained for maximum likelihood factor analysis.

AIC and BIC Comparison

To provide a more rigorous statistical basis for factor retention, Akaike's Information Criterion (AIC) and the Bayesian Information Criterion (BIC) were computed for two-, three-, and four-factor maximum likelihood solutions. The model with the lowest AIC/BIC value is preferred, balancing goodness-of-fit against model complexity. Delta scores (differences relative to the best-fitting model) were calculated to assess the relative support for competing models, with $\Delta < 2$ indicating substantial support and $\Delta > 10$ suggesting the competing model is very unlikely. Both AIC and BIC point to the two factor model as the best fitting one. Delta scores were therefore computed for the three and four factor models against the two factor one. The following table presents the results from this analysis. All numbers are rounded to the second decimal point.

Model	AIC	BIC	Delta AIC	Delta BIC
2 factor	3.85	11.26	-	-
3 factor	5.85	16.97	2.00	5.70
4 factor	7.85	22.68	4.00	11.41

Final Decision

While the AIC/BIC comparison favours a more parsimonious two-factor solution, several considerations led to the retention of three factors for the final analysis:

1. Theoretical interpretability: The three-factor structure directly corresponds to the expected cognitive domains (visual-spatial, verbal, processing speed), which is valuable for educational assessment.
2. Variance explained: The three-factor model explains 54% of the variance versus 48% for two factors, a meaningful practical difference.

3. Residual fit: As shown in later tasks, the three-factor model produces acceptable residuals (Frobenius norm = 0.163).
4. Expert judgement: In psychometric contexts, a theoretically grounded solution is often preferred over purely statistical heuristics, especially when information criteria disagree modestly (ΔAIC = 2).

Thus, the three-factor solution was adopted as the primary model for subsequent extraction, rotation, and interpretation.

Analysis and Model Fitting

The three-factor maximum likelihood solution (fa3) produced the following results.

SS loadings: Factor1 = 2.185, Factor2 = 1.343, Factor3 = 1.327 **Proportion variance explained:** Factor1 = 0.243, Factor2 = 0.149, Factor3 = 0.147 **Cumulative variance explained:** 0.539 (54%)

Goodness-of-fit test: $\chi^2(12) = 22.38$, $p = 0.0335$. The null hypothesis that 3 factors are sufficient is rejected at $\alpha = 0.05$ but not at $\alpha = 0.01$. Given the scree plot, theoretical interpretability, and the modest improvement with 4 factors, the 3-factor solution was retained.

Comparison with 4-factor model (fa4**):** The 4-factor solution explained 58% cumulative variance (only 4% additional). The 4th factor had an SS loading of 0.238 (negligible). The fit improved ($\chi^2(6) = 5.58$, $p = 0.472$), but parsimony favoured the 3-factor model.

Rotated Solutions

Both varimax (orthogonal) and promax (oblique) rotations were applied to the 3-factor solution.

Interpretation after varimax:

- **Factor1 (Spatial/Verbal split?):** Actually loads x4, x5, x6 → **Verbal ability**
- **Factor2:** Loads x1, x2, x3 → **Visual-spatial ability**

- **Factor3:** Loads x7, x8, x9 (x9 also loads on Factor2 modestly) → **Processing speed**

Oblique rotation (promax): The factor correlation matrix showed modest correlations between factors: Factor1–Factor2 = 0.25, Factor1–Factor3 = 0.26, Factor2–Factor3 = 0.13. The oblique solution produced a slightly cleaner loading pattern (x9 loaded more cleanly on Factor3). Given the modest factor correlations, the orthogonal (varimax) solution is reported for simplicity, though the oblique solution is theoretically plausible.

Final factor naming (based on varimax loadings):

- **Factor 1: Textual/Verbal ability** (x4, x5, x6)
- **Factor 2: Visual/Spatial ability** (x1, x2, x3)
- **Factor 3: Processing speed** (x7, x8, x9)

Model Adequacy: Residuals

The residual correlation matrix (observed minus reproduced) showed small residuals, mostly between -0.03 and 0.04. The Frobenius norm of the residual matrix was **0.163**, indicating a good fit (values below 0.20 are generally considered acceptable). The largest absolute residual was 0.042 (x7 with x4), suggesting that the 3-factor model slightly underestimates the relationship between processing speed and verbal ability. However, this was not considered problematic.

Factor Scores and Group Comparisons

Factor scores were computed using the principal component method and added to the dataset as pc1 (verbal), pc2 (spatial), and pc3 (processing speed). Mean scores were compared across schools, grades, and sex.

Interpretation of group differences:

- School: Pasteur students scored higher on spatial ability (pc2 = 0.33 vs. -0.36). Grant-White students scored higher on verbal ability (pc1 = 0.12 vs. -0.11). Differences on processing speed were negligible.
- Grade: Eighth graders outperformed seventh graders on verbal and spatial factors. Seventh graders scored slightly higher on processing speed (0.12 vs. -0.13).
- Sex: Males scored higher on processing speed (0.22 vs. -0.21). Females scored slightly higher on verbal ability (-0.07 vs. 0.07). Differences on spatial ability were negligible.

5. Discussion

The three-factor solution (verbal, spatial, processing speed) explains 54% of the variance and fits the data adequately (residual Frobenius norm = 0.163). The fit test is marginal ($p = 0.0335$), but the four-factor model adds only 4% variance with a negligible fourth factor. Parsimony favours three factors. Although AIC and BIC favoured an even more parsimonious two-factor model (ΔAIC = 2.00, ΔBIC = 5.71 for the three-factor solution), the three-factor structure was retained due to its alignment with established cognitive domains and the modest improvement in variance explained (54% vs. 48%).

Rotation (varimax) produced clean loadings: x4–x6 load on Factor1 (verbal), x1–x3 on Factor2 (spatial), x7–x9 on Factor3 (processing speed). Oblique rotation showed modest factor correlations (0.13–0.26), supporting distinct but related dimensions.

Factor scores revealed descriptive group differences:

- School: Pasteur higher on spatial; Grant-White higher on verbal.
- Grade: 8th graders higher on verbal and spatial; 7th graders higher on processing speed.
- Sex: Males higher on processing speed; females higher on verbal (small differences).

Limitations: ML assumes normality; no inferential testing on factor scores; data from 1939. A limitation of the factor retention decision is the reliance on multiple criteria that did not fully converge. Future research could cross-validate the factor structure on an independent sample or employ parallel analysis for additional guidance.

Conclusion: Three factors adequately summarise the nine test scores. The model is interpretable and useful for comparing student groups.

6. References

Multivariate Data Analysis Notes, Loukia Meligkotsidou

Multivariate Statistical Analysis, Dimitris Karlis

Methods of Multivariate Analysis, Rencher and Christensen, Wiley.

Appendix

The following three 3D scatterplots visualise the factor loadings of the

nine variables on the three retained factors under different rotation methods.

- Plot 1 (blue): Unrotated loadings. The initial solution shows a general factor pattern with variables clustered but not clearly separated along the axes.
- Plot 2 (violet): Varimax rotation (orthogonal). Loadings are rotated to simplify columns (factors). The clusters become more distinct, aiding interpretation of each factor as a separate latent dimension (verbal, spatial, speed).
- Plot 3 (sienna): Promax rotation (oblique). An oblique rotation allows factors to be correlated. The loading pattern is broadly similar to varimax, confirming the three-factor structure regardless of the rotation method chosen.

All three plots show three clear clusters of variables, each cluster corresponding to one of the expected domains (textual/verbal, visual/spatial, processing speed). Rotation improves interpretability without fundamentally altering the underlying structure.